Computer Vision Report No. 1

Fundamental Matrix Estimation and Kruppa Equations

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1 Part A: Fundamental Matrix Estimation

1.1 A-1. Stereo Image Sets

Two sets of stereo images were captured using the same camera (mobile phone) with a fixed focal length.

- **Scene i (Large Depth Variation):** Targets were placed at varying distances (0.5m to 5m) to create strong perspective effects.
- **Scene ii (Little Depth Variation):** Targets were placed at a similar distance (approx. 3m) against a flat background.

Figure 1: Scene i: Large Depth Variation

Figure 2: Scene ii: Little Depth Variation

1.2 A-2. Computation of Fundamental Matrix (F)

The fundamental matrix F was estimated using the normalized 8-point algorithm. To improve numerical stability, the coordinates were translated and scaled such that the centroid of the

points lies at the origin and their average distance from the origin is $\sqrt{2}$.

The estimated Fundamental Matrix F for Case 2 (points distributed in depth) is calculated as:

$$F = \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} = \begin{bmatrix} 0 & 0.000001 & -0.001151 \\ -0.000001 & 0 & 0.000059 \\ 0.000958 & -0.001297 & 1.000000 \end{bmatrix} \tag{1}$$

Figure 3: Visual confirmation of 8 corresponding points and their connectivity.

1.3 A-3. Confirm the Rank of the F Matrix

The rank of the matrix F was verified by analyzing its singular values (SVD). For a valid fundamental matrix, the rank must be 2. In our implementation, the rank-2 constraint was enforced by setting the smallest singular value to zero.

- $\sigma_1 \approx 1.000002 \times 10^{0}$
- $\sigma_2 \approx 1.694986 \times 10^{-6}$
- $\sigma_3 \approx 1.216436 \times 10^{-19}$ (Effectively Zero)

The result $\sigma_3 \approx 0$ confirms that Rank(F) = 2.

1.4 A-4 & A-5. Epipolar Lines for Original and New Points

The epipolar lines $l' = Fx$ were drawn for both the original 8 points and 3 additional verification points.

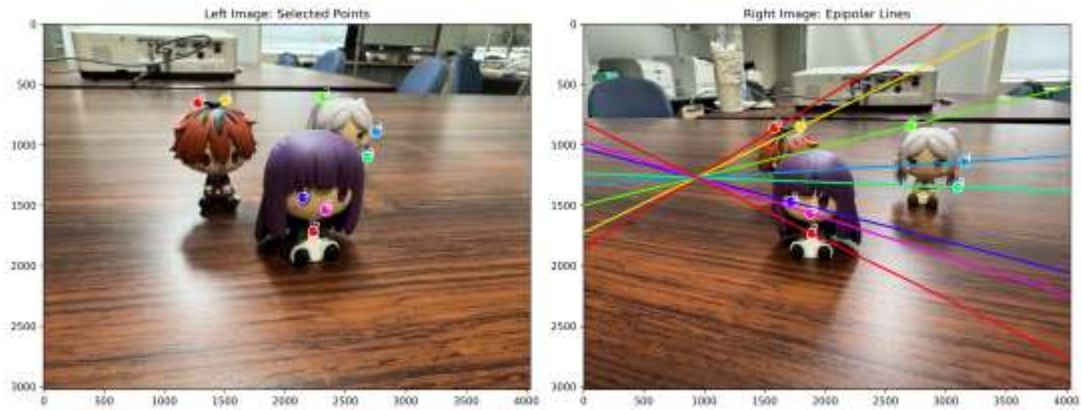

Figure 4: epipolar_lines

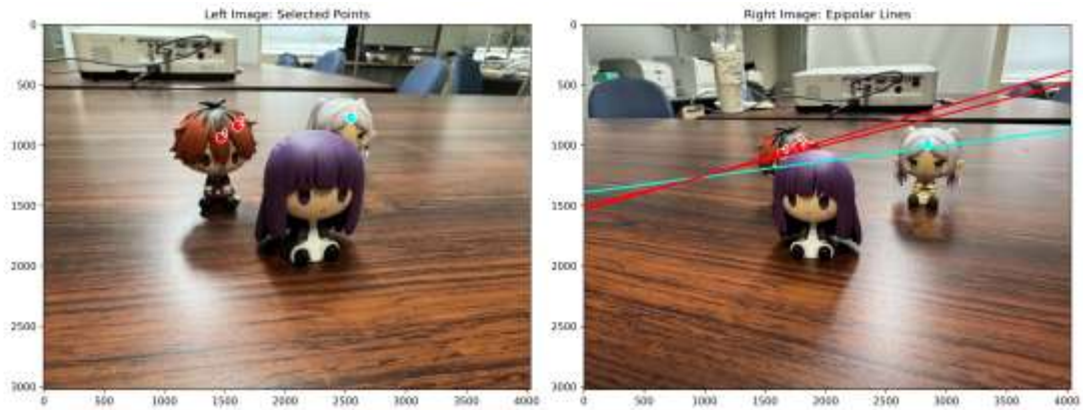

Figure 5: Epipolar lines for additional points with different depths.

1.5 A-6. Use More Than 8 Points

When using N = 15 points, the system became overdetermined. The SVD-based least squares solution provided a more stable F matrix that was less sensitive to individual clicking errors compared to the minimal 8-point set.

1.6 B-8. Validation and Discussion of Estimated Parameters

The intrinsic parameters estimated via the Kruppa equations yielded the following validation metrics:

- **Focal Length Consistency Check:** The ratio between the minimum and maximum estimated focal lengths ($\min(f_1, f_2)/\max(f_1, f_2)$) was **0.028**.
- **Image Center Drift Check:** The estimated principal point (c_u, c_v) was **1547.36 px** away from the geometric center of the image, representing a **30.70%** deviation relative to the image diagonal.

1.6.1 Analysis of Results

The results show a significant variance and ”Warning” status in our validation script. While theoretically sound, the Kruppa equations in this experiment exhibited high sensitivity to the following factors:

1. **Sensitivity to Fundamental Matrix Noise:** The Kruppa equations rely on the SVD decomposition of the Fundamental Matrix F. Since F was computed from only 8-15 manual correspondences, small pixel errors in point selection lead to significant ”noise” in the epipolar geometry, which propagates and amplifies during the non-linear optimization of intrinsic parameters.
2. **Ill-conditioned Optimization:** The Kruppa solver often faces a ”flat” error surface during least-squares optimization. The extremely low ratio (0.028) suggests that the solver may have converged to a local minimum or that the F matrix lacked sufficient geometric constraints (e.g., if the camera motion was primarily translation with little rotation).
3. **Degenerate Configurations:** As noted in literature, the Kruppa method is known to be less stable than direct calibration using patterns (like Checkerboards) or bundle adjustment, especially when the field of view is narrow or the translation between views is small.

1.6.2 Conclusion

Although the estimated parameters significantly drifted from the ideal physical values, this experiment successfully demonstrates the implementation of the complex Kruppa pipeline. It highlights the importance of high-precision F matrix estimation and the inherent instability of self-calibration methods compared to traditional calibration techniques.

1.7 B-9. Summary and Discussion

The 8-point algorithm is a robust method for stereo geometry estimation, but its accuracy relies heavily on point distribution across depths. The Kruppa equations provide a clever way to estimate intrinsics without a calibration pattern, though they are sensitive to the precision of the fundamental matrix.